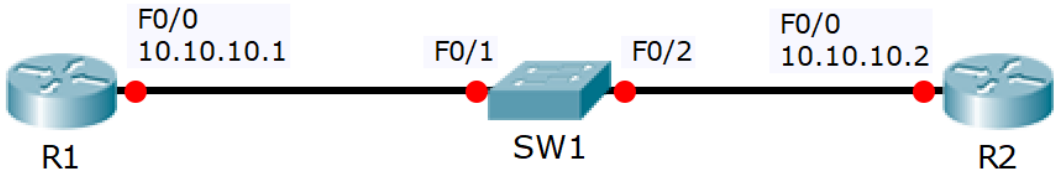
14 Cisco Router and Switch Basics - Lab Exercise

In this lab you will complete a basic configuration on a switch, verify Cisco Discovery Protocol CDP and analyse the effects of interface speed and duplex configuration.

Lab Topology



Load the Startup Configurations

Download the '14 Cisco Router and Switch Basics.zip' file here. *Extract* the project .pkt file then open it in Packet Tracer. Do not try to open the project from directly inside the zip file.

Cisco Router and Switch Initial Configuration

1) Configure Router 1 with the hostname 'R1'

2) Configure Router 2 with the hostname 'R2'

3) Configure Switch 1 with the hostname 'SW1'

4) Configure the IP address on R1 according to the topology diagram

5) Configure the IP address on R2 according to the topology diagram

6) Give SW1 the management IP address 10.10.10.10/24

7) The switch should have connectivity to other IP subnets via R2

8) Verify the switch can ping its default gateway

9) Enter suitable descriptions on the interfaces connecting the devices

10) On SW1, verify that speed and duplex are automatically negotiated to 100 Mbps full duplex on the link to R1

11) Manually configure full duplex and FastEthernet speed on the link to R2

12) What version of IOS is the switch running?

CDP Configuration

13) Verify the directly attached Cisco neighbors using Cisco Discovery Protocol

14) Prevent R1 from discovering information about Switch 1 via CDP

15) Flush the CDP cache on R1 by entering the 'no cdp run' then 'cdp run' commands in global configuration mode

16) Verify that R1 cannot see SW1 via CDP

Switch Troubleshooting

17) Verify the status of the switch port connected to R2 with the `show ip interface brief` command. It should show status and protocol up/up.

18) Shut down the interface connected to R2 and issue a `show ip interface brief` command again. The status and protocol should show administratively down/down.

19) Bring the interface up again. Verify the speed and duplex setting.

20) Set the duplex to half on Switch 1. Leave the settings as they are on R2.

21) Verify the state of the interface.

22) Set the duplex back to full duplex.

23) Set the speed to 10 Mbps.

24) Check if the interface is still operational.

25) Check if the interface is operational on R2. What is the status of the interface?

14 Cisco Router and Switch Basics - Answer Key

In this lab you will complete a basic configuration on a switch, verify Cisco Discovery Protocol CDP and analyse the effects of interface speed and duplex configuration.

Cisco Router and Switch Initial Configuration

1) Configure Router 1 with the hostname ‘R1’

```
Router(config)#hostname R1
R1(config)#
```

2) Configure Router 2 with the hostname ‘R2’

```
Router(config)#hostname R2
R2(config)#
```

3) Configure Switch 1 with the hostname ‘SW1’

```
Switch(config)#hostname SW1
SW1(config)#
```

4) Configure the IP address on R1 according to the topology diagram

```
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 10.10.10.1 255.255.255.0
R1(config-if)#no shutdown
```

5) Configure the IP address on R2 according to the topology diagram

```
R2(config)#interface FastEthernet0/0
R2(config-if)#ip address 10.10.10.2 255.255.255.0
R1(config-if)#no shutdown
```

6) Give SW1 the management IP address 10.10.10.10/24

```
SW1(config)#interface vlan1
SW1(config-if)#ip address 10.10.10.10 255.255.255.0
SW1(config-if)#no shutdown
```

7) The switch should have connectivity to other IP subnets via R2

```
SW1(config)#ip default-gateway 10.10.10.2
```

8) Verify the switch can ping its default gateway

```
SW1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/2/8 ms
```

9) Enter suitable descriptions on the interfaces connecting the devices

```
R1(config)#interface FastEthernet 0/0
R1(config-if)#description Link to SW1

R2(config-if)#interface FastEthernet 0/0
R2(config-if)#description Link to SW1

SW1(config)#interface FastEthernet 0/1
SW1(config-if)#description Link to R1
SW1(config-if)#interface FastEthernet 0/2
SW1(config-if)#description Link to R2
```

10) On SW1, verify that speed and duplex are automatically negotiated to 100 Mbps full duplex on the link to R1

```
SW1#show interface f0/1
FastEthernet0/1 is up, line protocol is up (connected)
Hardware is Lance, address is 00e0.8fd6.8901 (bia
00e0.8fd6.8901)
Description: Link to R1
BW 100000 Kbit, DLY 1000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s
```

11) Manually configure full duplex and FastEthernet speed on the link to R2

```
SW1(config)#interface FastEthernet 0/2
SW1(config-if)#speed 100
SW1(config-if)#duplex full
```

Don't forget to configure matching settings on R2!

```
R2(config)#interface FastEthernet 0/0
R2(config-if)#speed 100
R2(config-if)#duplex full
```

12) What version of IOS is the switch running?

```
SW1#show version
Cisco IOS Software, C2960 Software (C2960-LANBASE-M),
Version 12.2(25)FX, RELEASE SOFTWARE (fc1)
```

CDP Configuration

13) Verify the directly attached Cisco neighbors using Cisco Discovery Protocol

```
SW1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID Local Intrfce Holdtme Capability Platform Port ID
R1        Fas 0/1       170      R         C2800  Fas 0/0
R2        Fas 0/2       134      R         C2800  Fas 0/0
```

14) Prevent R1 from discovering information about Switch 1 via CDP

```
SW1(config)#interface FastEthernet 0/1
SW1(config-if)#no cdp enable
```

15) Flush the CDP cache on R1 by entering the 'no cdp run' then 'cdp run' commands in global configuration mode

```
R1(config)#no cdp run
R1(config)#cdp run
```

16) Verify that R1 cannot see SW1 via CDP

```
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
R1#
```

Switch Troubleshooting

17) Verify the status of the switch port connected to R2 with the `show ip interface brief` command. It should show status and protocol up/up.

```
SW1#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
Vlan1                  10.10.10.10     YES manual up                    up
FastEthernet0/1        unassigned      YES unset  up                    up
FastEthernet0/2        unassigned      YES unset  up                    up
```

18) Shut down the interface connected to R2 and issue a `show ip interface brief` command again. The status and protocol should show administratively down/down.

```
SW1(config)#interface FastEthernet 0/2
SW1(config-if)#shutdown
*Mar  1 00:44:34.212: %LINK-5-CHANGED: Interface
FastEthernet0/2, changed state to administratively down
*Mar  1 00:44:35.219: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet0/2, changed state to down
```

```
SW1(config-if)#do show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
Vlan1                  10.10.10.10     YES manual up                    up
FastEthernet0/1        unassigned      YES unset  up                    up
FastEthernet0/2        unassigned      YES unset  administratively down down
```

19) Bring the interface up again. Verify the speed and duplex setting.

```
SW1(config)#interface FastEthernet 0/2
SW1(config-if)#no shutdown
SW1(config-if)#
*Mar  1 00:45:52.637: %LINK-3-UPDOWN: Interface
FastEthernet0/2, changed state to up
*Mar  1 00:45:53.644: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet0/2, changed state to up
```

```
SW1#sh interface f0/2
FastEthernet0/2 is up, line protocol is up (connected)
Hardware is Lance, address is 00e0.8fd6.8902 (bia
00e0.8fd6.8902)
BW 100000 Kbit, DLY 1000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s
```

20) Set the duplex to half on Switch 1. Leave the settings as they are on R2.

```
SW1(config-if)#duplex half
SW1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state
to down

     %LINEPROTO-5-UPDOWN: Line protocol on Interface
     FastEthernet0/2, changed state to down
```

21) Verify the state of the interface.

The interface is down/down. It will not forward traffic.

```
SW1#show ip interface brief
Interface      IP-Address OK? Method Status Protocol
FastEthernet0/1 unassigned YES manual up up
FastEthernet0/2 unassigned YES manual down down
```

22) Set the duplex back to full duplex.

```
SW1(config)#int f0/2
SW1(config-if)#duplex full
SW1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state
to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/2, changed state to up
```

23) Set the speed to 10 Mbps.

```
SW1(config)#int f0/2
SW1(config-if)#speed 10
SW1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state
to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/2, changed state to down
```

24) Check if the interface is still operational.

```
SW1#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
Vlan1                  10.10.10.10     YES manual up                    up
FastEthernet0/1        unassigned      YES unset  up                    up
FastEthernet0/2        unassigned      YES unset  down                  down
```

The interface status is down/down.

25) Check if the interface is operational on R2. What is the status of the interface?

```
R2#show ip interface brief
Interface                 IP-Address      OK? Method Status                Protocol
FastEthernet0/0           10.10.10.2      YES manual up                    down
```

The interface status is up/down on R2.